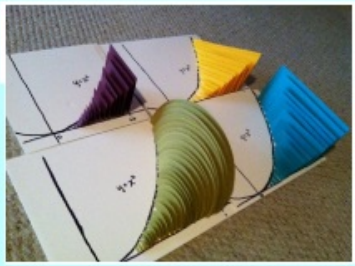
CHA-5.B.1 Volumes of solids with square and rectangular cross sections can be found using definite integrals and the area formulas for these shapes.
CHA-5.B.2 Volumes of solids with triangular cross sections can be found using definite integrals and the area formulas for these shapes.
CHA-5.B.3 Volumes of solids with semicircular and other geometrically defined cross sections can be found using definite integrals and the area formulas for these shapes.

8-7 & 8-8 Volume of Solids with Cross Sections

Previously, we have found the area bounded by curves and the volume generated by revolving a given region around a horizontal or vertical axis. Remember:

Volume of a Solid	The volume of a solid can always be found by integrating area: $\text{Volume} = \int_a^b A(x)\ dx \quad \text{or} \quad \int_c^d A(y)\ dy$

Today, we will look at finding volume Using known cross-sections.



A cross-section is simply a very small "slice" of a solid. If all these slices are the same shape throughout the solid, then we can use calculus to accumulate the cross sections and find the total volume.

In our examples, we will consider our bounded region to be the **BASE** of the solid. Our cross sections will **RISE UP** out of the base (see the illustration at the right).

How to find the Volume of a Solid with Cross-Sections
1.) Draw the region (this will be the base of your solid). 2.) Draw a line through the region (top to bottom or right to left) perpendicular to the x- or y-axis, as described in the problem. This line represents a piece of the cross-sectional shape. 3.) Determine the expression that represents the area of the cross-section. 4.) Use the Volume formula to find the volume.

The AP Calculus Exam will focus upon four familiar shapes from geometry: the square, the rectangle, the semi-circle and several varieties of triangles.

Geometric Shape	Required Information	Area Formula
Square	length of a side	$A = s^2$
Rectangle	length and width of two sides	$A = l \times w$
Semi-circle	radius	$A = \frac{1}{2}\pi r^2$
Triangle	length of base and height	$A = \frac{1}{2} b \times h$

The following chart contains area formulas you should know for the AP exam. If you choose not to memorize these, you must be able to derive them from the area formulas on the previous page.

Note: This chart assumes all cross sections are taken perpendicularly to the x-axis and the length from the top of the base to the bottom is defined in terms of x. If the cross-section is perpendicular to the y-axis, you will need the length from the right of the base to the left, defined in terms of y.

Cross-Sectional Shape	Known Information (on the base)	Area Formula
Square	SIDE	$A = (\text{side})^2$
Rectangle	LENGTH (aka BASE) OR WIDTH (aka HEIGHT)	$A = \text{LENGTH} \times \text{WIDTH}$ or $A = \text{BASE} \times \text{HEIGHT}$
Semicircle ①	DIAMETER	$A = \frac{\pi}{8}(\text{DIAMETER})^2$
Quarter-circle	RADIUS	$A = \frac{\pi}{4}(\text{RADIUS})^2$
Equilateral Triangle ②	SIDE	$A = \frac{\sqrt{3}}{4}(\text{SIDE})^2$
Isosceles Right Triangle (with Leg on the base)	LEG	$A = \frac{1}{2}(\text{LEG})^2$
Isosceles Right Triangle (with Hypotenuse on the base)	HYPOTENUSE	$A = \frac{1}{4}(\text{HYPOTENUSE})^2$

① Semi-circle

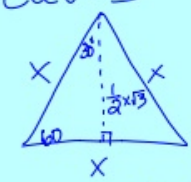
$r = \frac{1}{2}d$

$A = \frac{1}{2}\pi r^2$

$= \frac{1}{2}\pi\left(\frac{1}{2}d\right)^2$

$= \frac{1}{8}\pi d^2$

② EQU. Δ



$\sin 60° = \frac{h}{x}$

$\frac{\sqrt{3}}{2} = \frac{h}{x}$

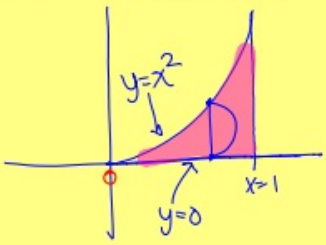
$x \cdot \frac{\sqrt{3}}{2} = h$

$\text{Area} = \frac{1}{2} b \cdot h$

$= \frac{1}{2}x \cdot \frac{1}{2}x\sqrt{3}$

$= \frac{\sqrt{3}}{4}x^2$

Example 1: Let R be the region bounded by $y = x^2$, $y = 0$, and $x = 1$. Find the volume of a solid S if its base is bounded by the region R and whose cross-sections perpendicular to the x-axis are semicircles.



$$\text{Area} = \frac{\pi}{8}\left[\text{diameter}\right]^2$$

$$\text{diameter} = x^2 - 0$$

$$V = \int_0^1 \frac{\pi}{8}\left[x^2 - 0\right]^2 dx$$

$$= \frac{\pi}{8}\int_0^1 x^4\, dx$$

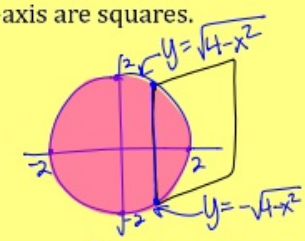
$$= \frac{\pi}{8}\left[\frac{1}{5}x^5\right]\Big|_0^1$$

$$= \frac{\pi}{8}\left[\frac{1}{5} - 0\right] = \boxed{\frac{\pi}{40}}$$

Example 2: Find the volume of the solid in which the base is a circle centered at the origin with radius 2 units whose cross sections are described.

$x^2 + y^2 = 4$

(a) The cross-sections perpendicular to the x-axis are squares.



$x^2+y^2=4$

$y^2=4-x^2$

$y=\pm\sqrt{4-x^2}$

$\text{Area} = [\text{side}]^2$

$\text{Side} = \sqrt{4-x^2} - -\sqrt{4-x^2}$

$$V = \int_{-2}^{2} \left[2\sqrt{4-x^2}\right]^2 dx$$

$$= \int_{-2}^{2} 4(4-x^2)\, dx$$

$$= 4\int_{-2}^{2} (4-x^2)\, dx$$

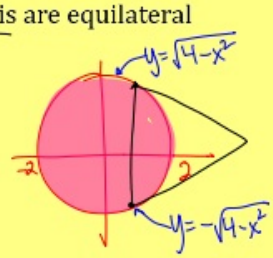
$$= 4\left[4x - \tfrac{1}{3}x^3\right]\Big|_{-2}^{2}$$

$$= 4\left(8 - \tfrac{8}{3} - \left(-8 + \tfrac{8}{3}\right)\right)$$

$$= \boxed{\frac{128}{3}}$$

Example 2: Find the volume of the solid in which the base is a circle centered at the origin with radius 2 units whose cross sections are described.

(b) The cross-sections perpendicular to the x-axis are equilateral triangles



$$A = \frac{\sqrt{3}}{4}\left[\text{Side}\right]^2$$

$$\text{Side} = \sqrt{4-x^2} - -\sqrt{4-x^2}$$

$$V = \int_{-2}^{2} \frac{\sqrt{3}}{4}\left[2\sqrt{4-x^2}\right]^2 dx$$

$$= \frac{\sqrt{3}}{4}\int_{-2}^{2} 4(4-x^2)\,dx$$

$$= \sqrt{3}\int_{-2}^{2}(4-x^2)\,dx$$

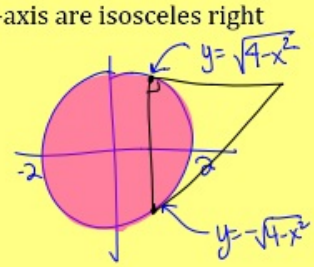
$$= \sqrt{3}\left[4x - \frac{1}{3}x^3\right]\Big|_{-2}^{2}$$

$$= \sqrt{3}\left(8 - \frac{8}{3} - \left(-8 + \frac{8}{3}\right)\right)$$

$$= \boxed{\frac{32\sqrt{3}}{3}}$$

Example 2: Find the volume of the solid in which the base is a circle centered at the origin with radius 2 units whose cross sections are described.

(c) The cross-sections perpendicular to the x-axis are isosceles right triangles with a leg on the base.



$$A=\frac{1}{2}[\text{leg}]^2$$

$$\text{Leg}=\sqrt{4-x^2}-\left(-\sqrt{4-x^2}\right)$$

$$V=\int_{-2}^{2}\frac{1}{2}\left[2\sqrt{4-x^2}\right]^2dx$$

$$=\frac{1}{2}\int_{-2}^{2}4(4-x^2)\,dx$$

$$=2\int_{-2}^{2}4-x^2\,dx$$

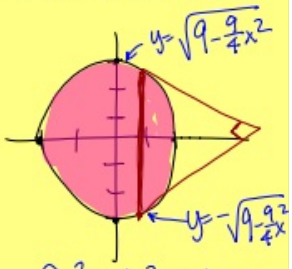
$$=2\left[4x-\frac{1}{3}x^3\right]\Big|_{-2}^{2}$$

$$=2\left(8-\frac{8}{3}-\left(-8+\frac{8}{3}\right)\right)=\boxed{\frac{64}{3}}$$

Example 3: The base of S is an elliptical region with boundary curve $9x^2 + 4y^2 = 36$. Cross-sections perpendicular to the x-axis are isosceles right triangles with hypotenuse on the base. Find the volume of S.

$$\frac{9x^2}{36} + \frac{4y^2}{36} = \frac{36}{36}$$

$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$



$$9x^2 + 4y^2 = 36$$
$$4y^2 = 36 - 9x^2$$
$$y^2 = 9 - \frac{9}{4}x^2$$
$$y = \pm\sqrt{9 - \frac{9}{4}x^2}$$

$$\text{Area} = \frac{1}{4}[\text{hypotenuse}]^2$$

$$\text{hypotenuse} = \sqrt{9 - \frac{9}{4}x^2} - -\sqrt{9 - \frac{9}{4}x^2}$$

$$V = \int_{-2}^{2} \frac{1}{4}\left[2\sqrt{9 - \frac{9}{4}x^2}\right]^2 dx$$

$$= \frac{1}{4}\int_{-2}^{2} 4\left(9 - \frac{9}{4}x^2\right) dx$$

$$= \frac{1}{4}\int_{-2}^{2} (36 - 9x^2)\, dy$$

$$= \frac{1}{4}\left[36x - 3x^3\right]\Big|_{-2}^{2}$$

$$= \frac{1}{4}\left[72 - 24 - (-72 + 24)\right]$$

$$= \boxed{24}$$

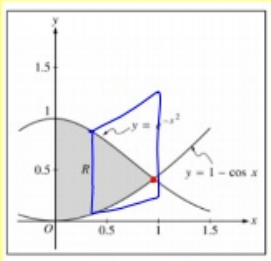
Example 4: (2000 AB1) Let R be the shaded region in the first quadrant enclosed by the graphs of $y = e^{-x^2}$, $y = 1 - \cos x$, and the y-axis as shown in the figure below.

(c) The region R is the base of a solid. For this solid, each cross section perpendicular to the x-axis is a square. Find the volume of this solid.

$\text{Area} = [\text{side}]^2$

$\text{Side} = e^{-x^2} - (1 - \cos x)$

$V = \int_0^A \left[e^{-x^2} - (1 - \cos x)\right]^2 dx$



X-value of pt. of int.

$x = .9419440\ldots = A$ (stored in calc.)